XRPL Validator — Daily Engineering Report · 2026-07-27

Period: 2026-07-27 00:00 to 24:00 US Eastern (2026-07-27 04:00 to 2026-07-28 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04); data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-07-27): 1.00000 (0 missed of 21,630). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(xrpl-ledger): reap a book level's dead offers before consulting the pool** (`01f09ba`, xrpl-validator) rippled steps the offer stream BEFORE it asks the AMM, and every dead offer stepped past is removed whether or not anything crossed: if (offers.step()) // BookStep.cpp:835 if (tryAMM(offers.tip().quality())) // anchor = first LIVE offer do { ... } while (offers.step()); else tryAMM(std::nullopt); // exhausted -> maxOffer return {offers.permToRemove(), ...}; // BookStep.cpp:852
- **fix(xrpl-ledger): step past a satisfied fill, and reap dust-backed offers** (`130c246`, xrpl-validator) Two more of `OfferStream::step`'s removal semantics, both needed to close #105778999 6B2A11B3 (5v15, missing 10, extra 0) -- a circular self-pay, SendMax 140 XRP into POSAA, tfPartialPayment.
- **fix(xrpl-ledger): carry the trailing reap across book levels** (`ba73d4f`, xrpl-validator) The trailing step added in 130c246 stopped at the end of the page chain of the level it had been crossing on. rippled's stream spans the whole BOOK: `OfferStream::step` carries no quality test at all, so after a consumed offer it keeps advancing — and reaping — across quality levels until it reaches a LIVE offer, and only then does `execOffer`'s `checkQualityThreshold` return false and end the walk …
- **fix(xrpl-ledger): a crossing that exhausts the account rests no remainder** (`bb522bf`, xrpl-validator) "If offer crossing exhausted the account's funds don't create the offer" (OfferCreate.cpp:479-484). This is the companion to the pre-crossing `accountFunds` clamp from 1de6d87: that one bounds what may be SPENT, this one bounds what may REST.
- **fix(xrpl-ledger): tick re-derivation divides with the STAmount rule** (`1edaa63`, xrpl-validator) #105777146 9BA91A9E sells 0.00059094 WETH for 1 XRP. The WETH issuer publishes TickSize 6, so the rate rounds up to

6 significant digits and the non-exact side is re-derived before placing. We derived TakerGets 0.000590939712330548 where mainnet stored 0.0005909397123305481 -- one ulp higher -- and that single ulp moved the rested offer to book level ...ABFB5801 where mainnet wrote ...ABFB5800. 4v4, the ledger's …

- **fix(xrpl-ledger): a bridged pass needs its book composition within the limit** (`625399c`, xrpl-validator) #105807256 84FD7DC8, the last divergence in the harness. A tfSell|tfPassive OfferCreate selling 3105.662739 BBRL for 607.478737 RLUSD: mainnet rests it in 4 nodes, we crossed 8 objects (12v4, all extra, nothing missing).
- **fix(xrpl-ledger): a frozen holder can fund nothing** (`1c46cda`, xrpl-validator) rippled reads spendable IOU through `accountHolds` with `fhZERO_IF_FROZEN`, so a holder whose line the issuer has frozen has NO funds at all, however large the balance. `isFrozen` (RippleStateHelpers.cpp:127) is the issuer's GLOBAL freeze, or the ISSUER's side of the line -- `(issuer > account) ? lsfHighFreeze : lsfLowFreeze`. Our `available()` read the balance and nothing else, so a frozen holder could trade freely.

Docs and operations

- **docs(xrpl-ledger): correct the stale directory.rs scope note** (`51f6fef`, xrpl-validator)

Independence program

No independence record for UTC day 2026-07-27; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **WASM-Gated Escrow Finish Predicates** (hooks) Add flag tfWasmCondition and SField sfFinishPredicate (a size-capped, ≤4KB WASM blob run by a deterministic bounded interpreter — fixed step ceiling,
- **Threshold-Delayed Outbound Payment Holds** (custody) Add AccountPolicySet writing an AccountPolicy SLE (owner-reserved, keylet::accountPolicy(account)) with sfPolicyThreshold, sfHoldDelay (ledgers), and
- **Snapshot-Based MPT Corporate Actions** (rwa) Add issuer-only CorporateActionCreate, which creates an ltCORPORATE_ACTION object (owner-reserved, keylet::corpAction(issuer, sequence)) referencing a

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
07-27 17:06	—	xrpl-validator	`01f09ba`	fix(xrpl-ledger): reap a book level's dead offers before consulting the pool
07-27 17:07	—	xrpl-validator	`51f6fef`	docs(xrpl-ledger): correct the stale directory.rs scope note
07-27 23:22	—	xrpl-validator	`130c246`	fix(xrpl-ledger): step past a satisfied fill, and reap dust-backed offers
07-27 23:45	—	xrpl-validator	`ba73d4f`	fix(xrpl-ledger): carry the trailing reap across book levels
07-28 00:22	—	xrpl-validator	`bb522bf`	fix(xrpl-ledger): a crossing that exhausts the account rests no remainder
07-28 01:15	—	xrpl-validator	`1edaa63`	fix(xrpl-ledger): tick re-derivation divides with the STAmount rule
07-28 01:50	—	xrpl-validator	`625399c`	fix(xrpl-ledger): a bridged pass needs its book composition within the limit
07-28 03:18	—	xrpl-validator	`1c46cda`	fix(xrpl-ledger): a frozen holder can fund nothing

Times in UTC. A commit appears on the day it was committed to main or merged into it.